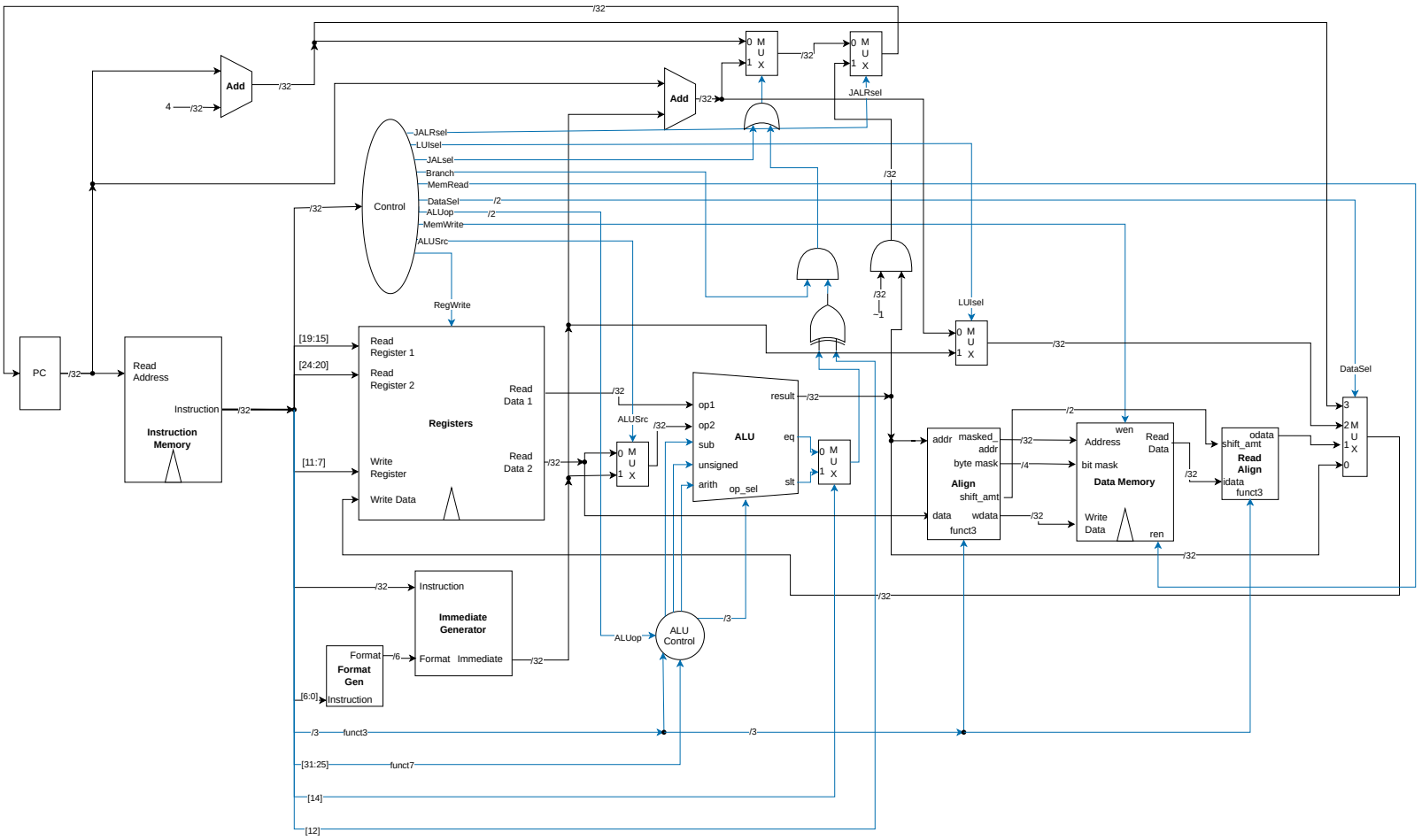
/32
Add
4
/32
/32
Add
/32
0 M U X 1
/32
0 M U X 1
JALRsel
JALRsel
LUIsel
JALsel
Branch
MemRead
DataSel
/2
ALUop
/2
MemWrite
ALUSrc
Control
/32
RegWrite
/32
/32
−1
LUIsel
0 M U X 1
/32
PC
/32
Read Address
Instruction
Instruction Memory
/32
[19:15]
[24:20]
[11:7]
Read Register 1
Read Register 2
Registers
Write Register
Write Data
Read Data 1
Read Data 2
/32
/32
ALUSrc
/32
0 M U X 1
op1
op2
sub
unsigned
arith
ALU
op_sel
result
/32
eq
slt
0 M U X 1
addr
masked addr
byte mask
Align
shift_amt
data
wdata
funct3
/32
/4
/32
/2
wen
Address
Read Data
bit mask
Data Memory
Write Data
ren
/32
odata
shift_amt
Read Align
idata
funct3
DataSel
3
2 M U X 1
0
/32
/32
Instruction
Immediate Generator
Format
Immediate
/6
/32
Format
Format Gen
Instruction
[6:0]
ALUop
ALU Control
/3
/3
/3
funct3
[31:25]
funct7
[14]
[12]